HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

10 December 2003

Time allowed 2 hours

Attempt all questions

Total Marks 60

Only approved calculators may be used.

NUMERICAL SOLUTIONS

1.

i	v	d	$i^{(4)}$	$d^{(12)}$
0.10	0.909091	0.090909	0.096455	0.094933

2. (a) £129.27
   (b) £113.17

3. (a) 17.531
   (b) 15.631
   (c) 75.31

4. Bookwork

5. (a) (i) £7,040.40
      (ii) £68,378.12
   (b) £10,274.13

6. £57.79

7. (a) Verify that $1000\ \ddot{s}_{\overline{3}|}^{0.1} = 3,641$.
   (b) The yield in 2002–3 is $i_1 = 0.05$. The yield in 2004 is $i_2 = 0.154956$. The LIRR is the solution of:

$$(1 + \text{LIRR})^3 = (1 + i_1)^2(1 + i_2)^1 \Rightarrow \text{LIRR} = 0.0839.$$

   (c) TWRR is given by:

$$\left(\frac{1,070}{1,000}.\frac{2,152.5}{2,070}.\frac{3,641}{3,152.5}\right)^{1/3} - 1 = 0.0872.$$

8. Cannot be answered on the basis of the current syllabus (now part of Financial Mathematics 2).